



数据挖掘

Data Mining

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廈門大學
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Python基础实验



Anaconda



实验任务



结果提交

1. Anaconda

■ Anaconda, 中文大蟒蛇, 是一个开源的Python发行版本, 其包含了conda、Python等180多个科学包及其依赖项结合

■ 下载: <https://www.anaconda.com/download>

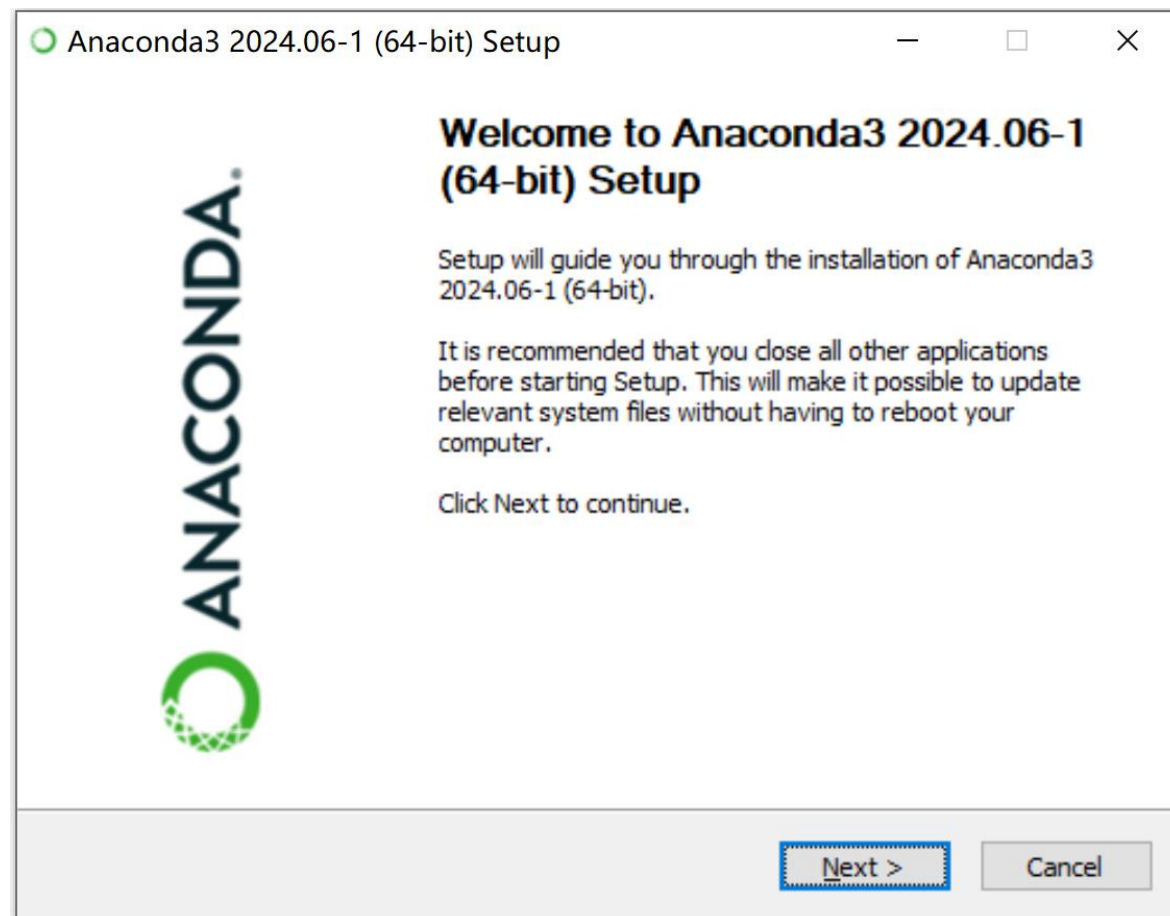
○ Anaconda3-2024.06-1-Windows-x86_64

■ 安装: <https://docs.anaconda.com/free/anaconda/install/>



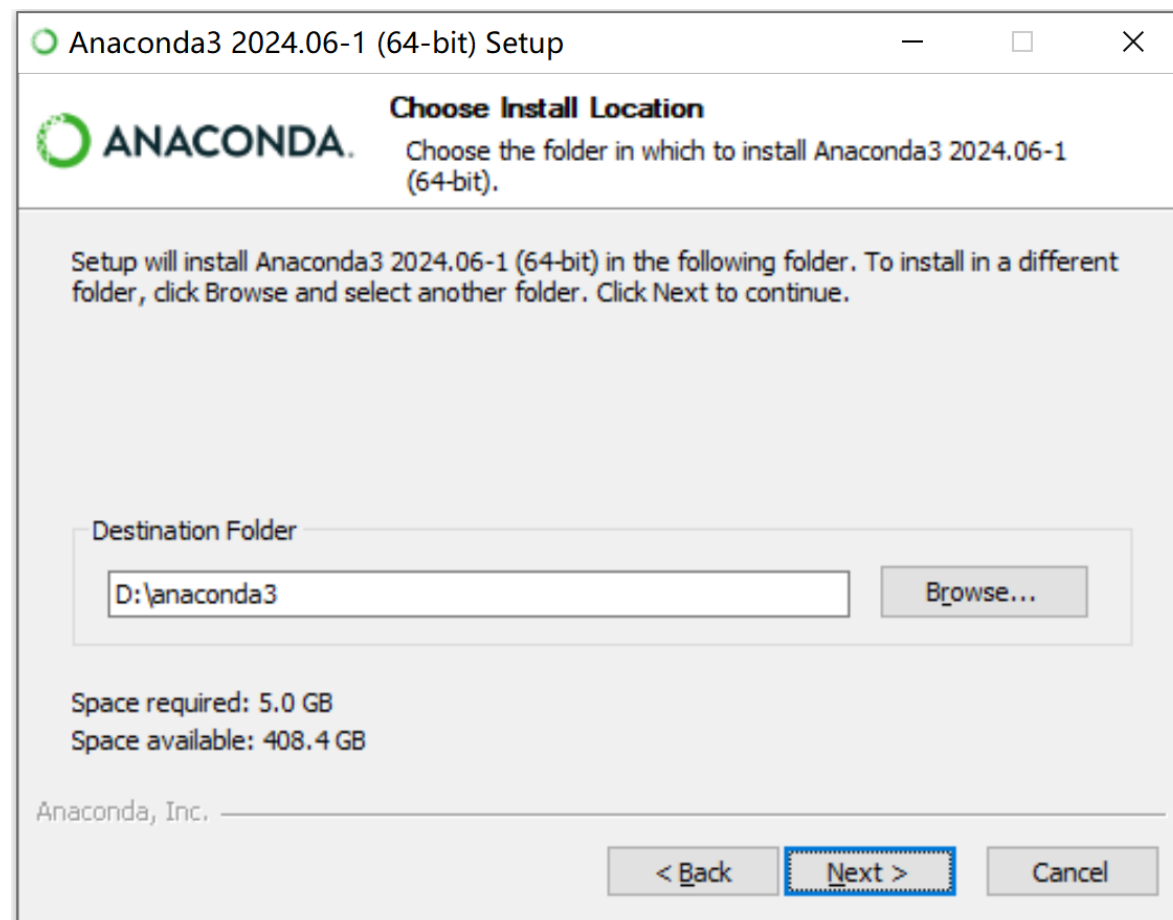
1. Anaconda

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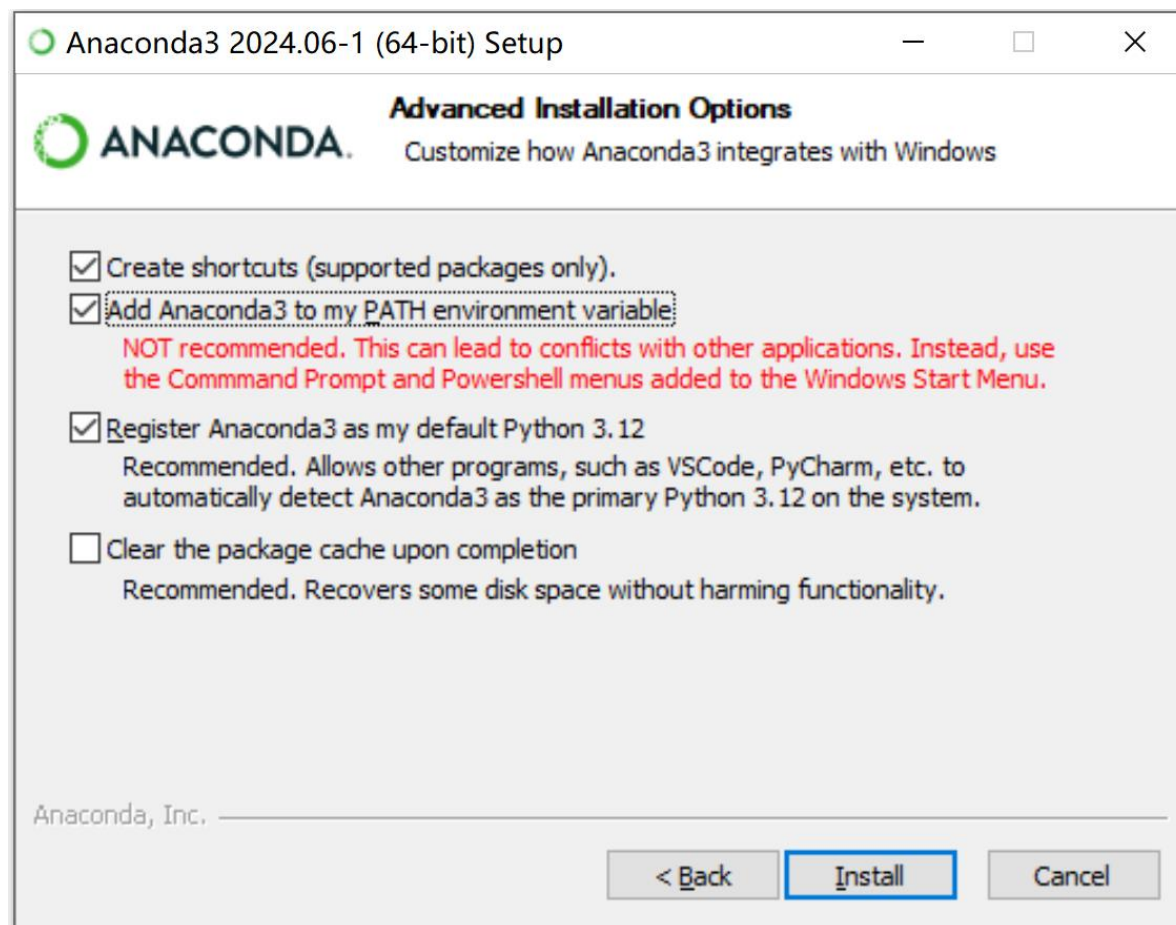
1. Anaconda

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1. Anaconda

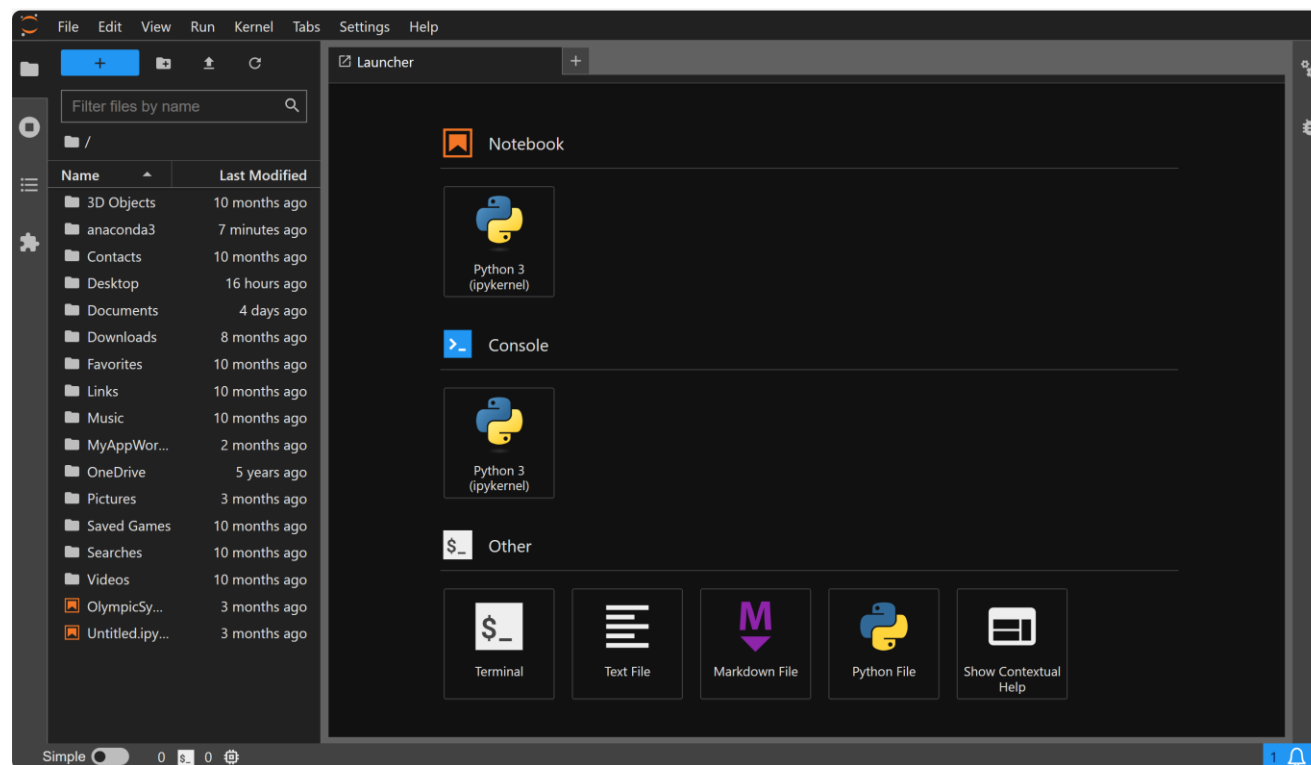
■ 安装: <https://docs.anaconda.com/free/anaconda/install/>



1. Anaconda

JupyterLab

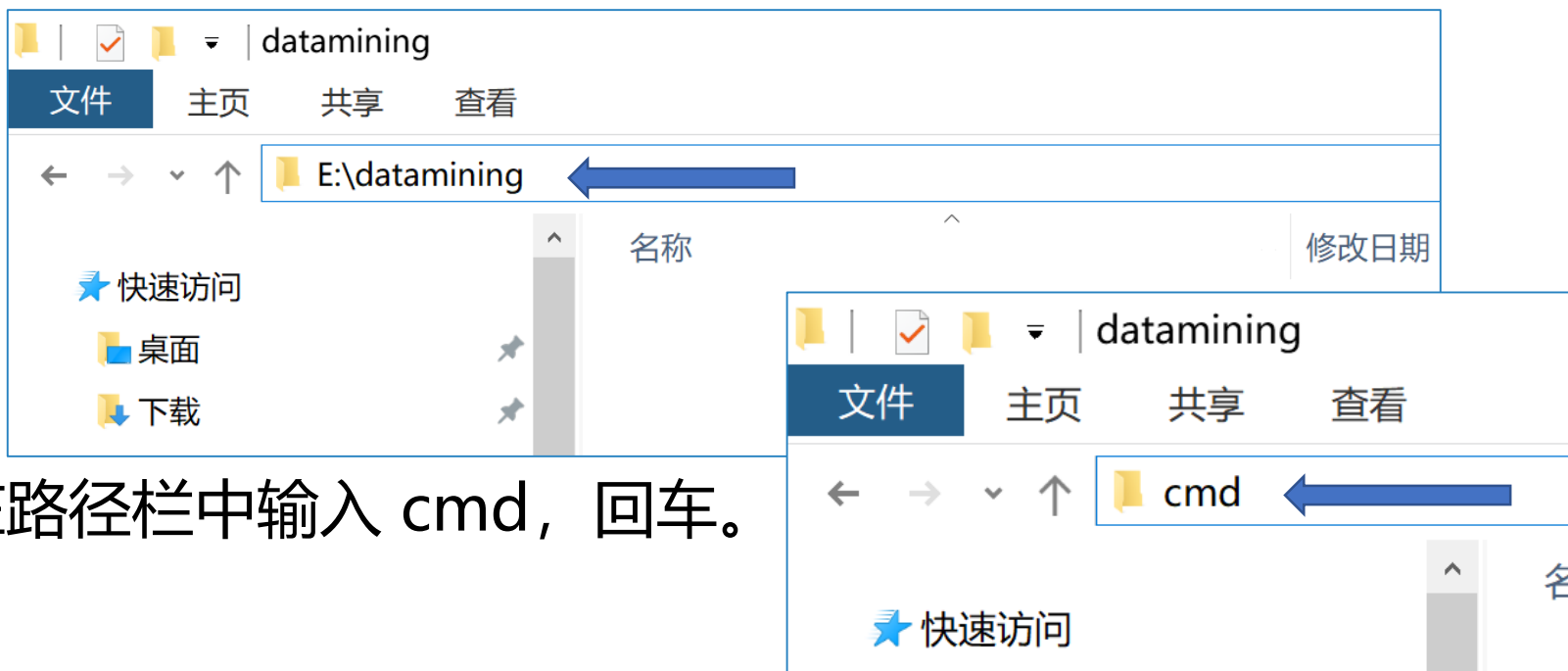
- JupyterLab 是 Jupyter Notebook 的下一代用户界面。它是一个基于 Web 的交互式开发环境，可以让用户以可视化的方式编辑、运行和分享代码。



1. Anaconda

JupyterLab

- 选定要存放Jupyter源文件的目录（比如：E:\datamining）。
- 在文件资源管理器中打开该目录。

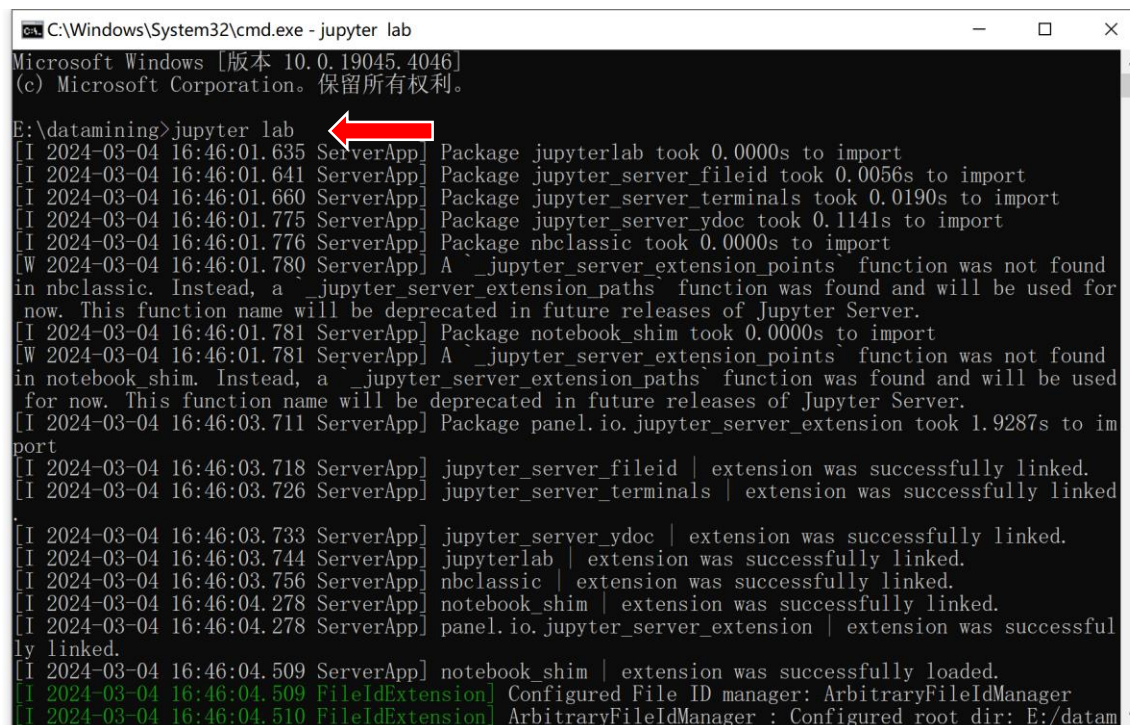


- 在路径栏中输入 cmd，回车。

1. Anaconda

JupyterLab

- 在弹出的界面光标处输入 `jupyter lab`，回车。
- 服务启动成功，**浏览器**中出现JupyterLab界面。



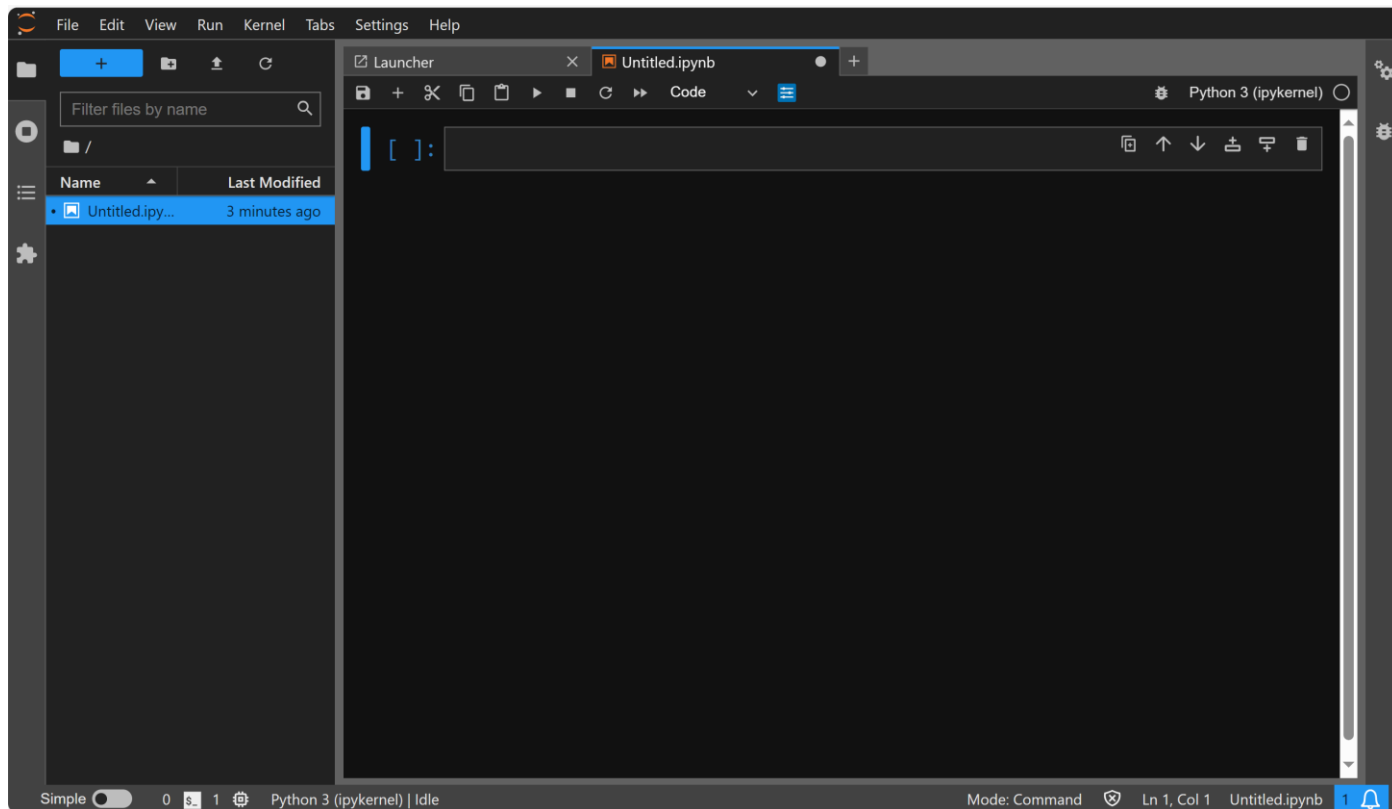
```
C:\Windows\System32\cmd.exe - jupyter lab
Microsoft Windows [版本 10.0.19045.4046]
(c) Microsoft Corporation. 保留所有权利。

E:\datamining>jupyter lab
[I 2024-03-04 16:46:01.635 ServerApp] Package jupyterlab took 0.0000s to import
[I 2024-03-04 16:46:01.641 ServerApp] Package jupyter_server_fileid took 0.0056s to import
[I 2024-03-04 16:46:01.660 ServerApp] Package jupyter_server_terminals took 0.0190s to import
[I 2024-03-04 16:46:01.775 ServerApp] Package jupyter_server_ydoc took 0.1141s to import
[I 2024-03-04 16:46:01.776 ServerApp] Package nbclassic took 0.0000s to import
[W 2024-03-04 16:46:01.780 ServerApp] A `_jupyter_server_extension_points` function was not found
in nbclassic. Instead, a `_jupyter_server_extension_paths` function was found and will be used for
now. This function name will be deprecated in future releases of Jupyter Server.
[I 2024-03-04 16:46:01.781 ServerApp] Package notebook_shim took 0.0000s to import
[W 2024-03-04 16:46:01.781 ServerApp] A `_jupyter_server_extension_points` function was not found
in notebook_shim. Instead, a `_jupyter_server_extension_paths` function was found and will be used
for now. This function name will be deprecated in future releases of Jupyter Server.
[I 2024-03-04 16:46:03.711 ServerApp] Package panel.io.jupyter_server_extension took 1.9287s to im
port
[I 2024-03-04 16:46:03.718 ServerApp] jupyter_server_fileid | extension was successfully linked.
[I 2024-03-04 16:46:03.726 ServerApp] jupyter_server_terminals | extension was successfully linked
.
[I 2024-03-04 16:46:03.733 ServerApp] jupyter_server_ydoc | extension was successfully linked.
[I 2024-03-04 16:46:03.744 ServerApp] jupyterlab | extension was successfully linked.
[I 2024-03-04 16:46:03.756 ServerApp] nbclassic | extension was successfully linked.
[I 2024-03-04 16:46:04.278 ServerApp] notebook_shim | extension was successfully linked.
[I 2024-03-04 16:46:04.278 ServerApp] panel.io.jupyter_server_extension | extension was successfu
ly linked.
[I 2024-03-04 16:46:04.509 ServerApp] notebook_shim | extension was successfully loaded.
[I 2024-03-04 16:46:04.509 FileIdExtension] Configured File ID manager: ArbitraryFileManager
[I 2024-03-04 16:46:04.510 FileIdExtension] ArbitraryFileManager : Configured root dir: E:/datam
```

1. Anaconda

JupyterLab

- 点击JupyterLab界面中的 Notebook 下的 Python 3图标。



1. Anaconda

JupyterLab

■ 在线帮助: <https://jupyterlab.readthedocs.io/en/stable/>

The screenshot displays the JupyterLab web application interface. On the left is a file browser sidebar with a search bar and a list of files and folders. The main area is divided into several panes:

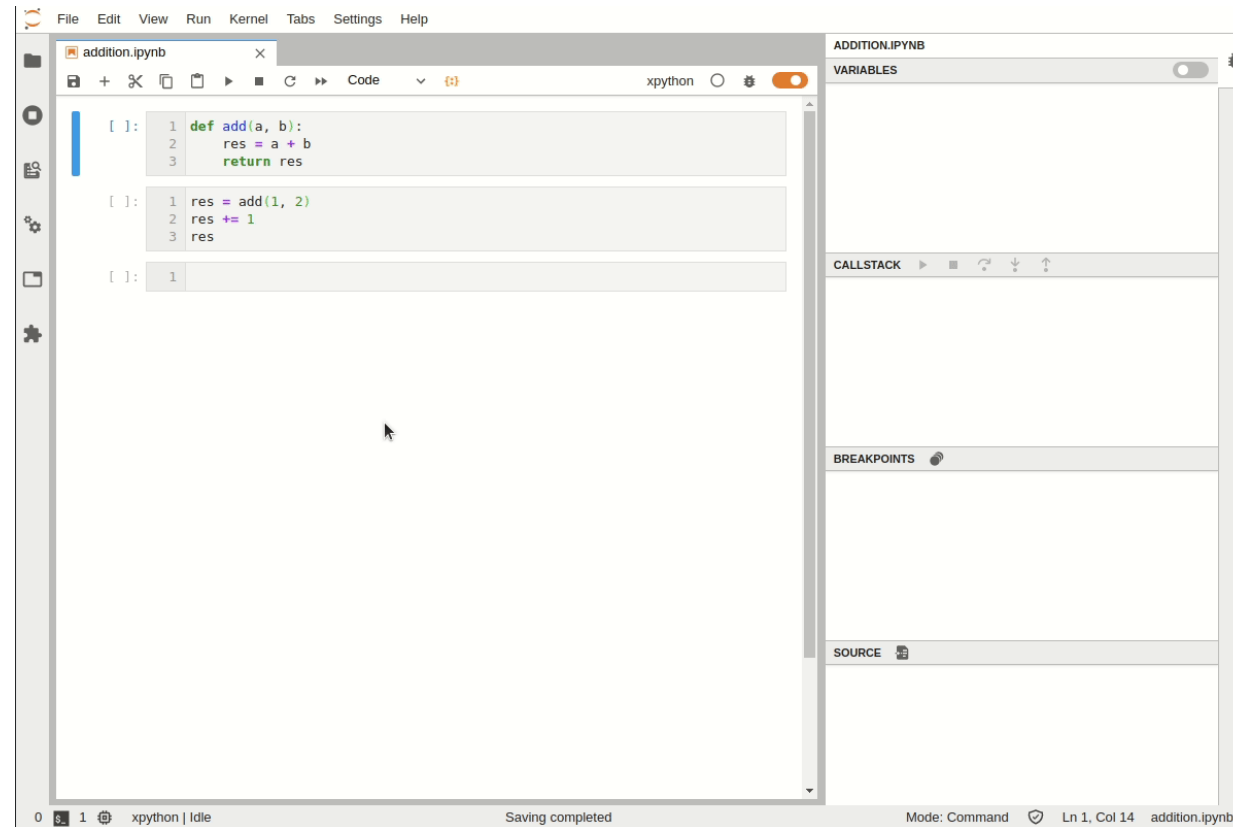
- Launcher:** Shows open tabs including 'README.md', 'Lorenz.ipynb', 'Terminal 1', 'Console 1', and 'Data.ipynb'.
- Code Editor:** Displays the code for 'Lorenz.ipynb'. It shows two code cells. The first cell imports the 'solve_lorenz' function and creates an interactive widget. The second cell shows the widget's output with sliders for 'sigma' (10.00), 'beta' (2.63), and 'rho' (28.00).
- Output View:** Shows the output of the code cells, including the same sliders and a 3D plot of the Lorenz attractor.
- lorenz.py:** A separate pane showing the source code for the 'solve_lorenz' function, which uses 'matplotlib' and 'scipy' to solve the Lorenz differential equations.

The bottom status bar indicates the current environment is 'Python' and shows the file path 'Ln 1, Col 1 Spaces: 4 lorenz.py 0'.

1. Anaconda

JupyterLab

■ Debug: <https://jupyterlab.readthedocs.io/en/stable/user/debugger.html>





Anaconda



实验任务



结果提交

2. 实验任务

自主学习

- 1.1 Python介绍
- 1.2 常见数据结构和基本语句
- 1.3 函数和模块
- 1.4 异常处理
- 1.5 文件读写

2. 实验任务

实验作业

- 完成 “实验1.ipynb” 中的实验作业



Anaconda



实验任务



结果提交

3. 结果提交

提交要求

- 在“实验1.ipynb”中将实验结果补充完整，配合适当的注释
- 在**学校课程平台**中提交该文件（需要压缩）
- 截至时间：**2025.09.17 23:59**
- 请注意：**不接收迟交作业！要独立完成，不允许抄袭！**